LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 10 Test (GSEB pattern): The Human Eye and the Colourful World

Chapter / Topic: The Human Eye and the Colourful World

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SECTION A

Q1. The human eye can focus on objects at different distances by adjusting the focal length of the eye lens. This is due to: **[1]**

Answer: (d) accommodation

Q2. Which part of the eye controls the size of the pupil? **[1]**

Answer: (a) Iris

Q3. The human eye forms the image of an object at its: **[1]**

Answer: (b) retina

Q4. The splitting of white light into its component colours is called: **[1]**

Answer: (c) dispersion

Q5. The least distance of distinct vision for a young adult with normal vision is about: **[1]**

Answer: (c) 25 cm

Q6. The clear sky looks blue because of: **[1]**

Answer: (a) scattering of light

Q7. The change in focal length of the eye lens is caused by the action of the: **[1]**

Answer: (d) ciliary muscles

Q8. A person cannot see objects beyond 2 m clearly. The defect is: **[1]**

Answer: (b) myopia

Q9. Myopia (near-sightedness) is corrected using a: **[1]**

Answer: (b) concave lens of suitable power

SECTION B

Q10. Why do stars twinkle? **[2]**

Answer: Starlight is refracted by the atmosphere, whose refractive index keeps changing (atmospheric refraction). Stars are very far away and act like point sources, so the apparent position of the star and the amount of light entering the eye keep changing. The star looks brighter at one moment and fainter the next.

- Atmospheric refraction with changing conditions: 1
- Flickering position/brightness: 1

Q11. What are the near point and the far point of a normal human eye? **[2]**

Answer: Near point: the closest point at which the eye can see an object clearly without strain, about 25 cm for a young adult. Far point: the farthest point it can see clearly, infinity for a normal eye.

- Near point 25 cm: 1
- Far point infinity: 1

Q12. What is dispersion of white light? What causes it? **[2]**

Answer: The splitting of white light into its seven component colours (VIBGYOR) by a prism. Different colours bend through different angles when they pass through the prism, so they separate.

- Definition: 1
- Different colours bend by different amounts: 1

SECTION C

Q13. A student can see distant objects clearly but cannot read a book held at a normal distance. **[3]**

(i) Hypermetropia (far-sightedness). [1]

(ii) The focal length of the eye lens is too long, or the eyeball has become too small. [1]

(iii) A convex lens of suitable power. It provides extra focusing power so that the image is formed on the retina. [1]

Q14. A person needs a lens of power −5.5 D to correct distant vision and a lens of power +1.5 D to correct near vision. **[3]**

(i) $f = 1/P = 1/(-5.5)$ m ≈ −0.18 m = −18.2 cm (concave lens) [1]

(ii) $f = 1/P = 1/1.5$ m ≈ +0.67 m = +66.7 cm (convex lens) [1]

(iii) A bifocal lens: the upper part concave (for distant vision) and the lower part convex (for near vision). [1]

SECTION D

Q15. A rainbow is a natural spectrum that appears in the sky after a rain shower. It is caused by the dispersion of sunlight by tiny water droplets in the… **[4]**

(i) Away from the Sun (with the Sun behind you). [1]

(ii) Tiny water droplets in the atmosphere. [1]

(iii) Refraction with dispersion, then internal reflection, then refraction again. [2]

- All three in order: 2 (two correct: 1)